

KyntriX: Simulation Beyond Limits — A Web-Based Robotics Simulation and Analysis Platform

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Abstract—KyntriX is a browser-based robotics simulation platform developed to combine robot modelling, kinematic analysis, interactive three-dimensional simulation, motion analysis, and practical robotics learning in one environment. The system provides 42 supported robot configurations, comprising industrial models and educational kinematic skeletons, and also supports construction of custom serial manipulators. Its workflow covers robot configuration, Denavit–Hartenberg (DH) modelling, forward and inverse kinematics, Jacobian and singularity analysis, trajectory generation, workspace exploration, real-time 3-D visualization, and simulator-linked practical challenges. A shared computational state connects the mathematical robot configuration with the rendered model, so analysis and visualization remain consistent throughout the workflow. The repository also contains validation suites and a documented 15-case audit of the Skeleton 6R model. The resulting platform is positioned as a practical browser-based laboratory for learning, testing, and experimenting with robotics.

Index Terms—robotics simulation, robot kinematics, Denavit–Hartenberg, forward kinematics, inverse kinematics, trajectory planning, 3-D visualization, robotics education

I. INTRODUCTION

Robotics education requires students to connect mathematical descriptions of a manipulator with its visible motion. DH parameters, homogeneous transformations, forward kinematics (FK), inverse kinematics (IK), Jacobians, trajectories, and workspace analysis are often introduced as separate topics. This can make it difficult for learners to see how a change in one quantity affects the complete robot configuration. Browser-based simulation offers a way to bring these ideas together while reducing the need for separate desktop tools.

Dey and Cheruvu presented a web-based graphical environment for three-dimensional robot modelling and kinematic study, including PUMA 560 analysis and CAD/STL-based visualization [1]. Peidró *et al.* demonstrated a browser-accessible virtual laboratory for visual kinematic analysis of five-bar and 3RRR planar parallel robots [2]. Nandi and Pattanaik proposed a web-based robotics simulator intended to make robot modelling and kinematic learning directly accessible through a browser [3]. These works establish the educational value of interactive robotics simulation while also motivating broader integration of robot models, analysis, and practice.

The project therefore focuses on a unified workflow rather than an isolated calculator or a visual-only simulator. A learner can select a robot, inspect its DH model, change joint values, observe the resulting 3-D motion, inspect transformations and end-effector pose, generate trajectories, study analysis

quantities, and use the same robot state in a practical challenge. The project presentation defines this workflow through six stages: robot model configuration, kinematic computation, trajectory generation, 3-D simulation and visualization, analysis and evaluation, and practical challenges.

II. PROBLEM STATEMENT AND OBJECTIVES

Students and researchers often move between different tools for robot simulation, kinematic calculation, workspace analysis, trajectory planning, and practical exercises. The project identifies four related gaps: limited robot support, separate analysis tools, a gap between computation and interactive simulation, and limited support for combining analysis with practical challenges. KyntriX is designed as a single browser-based environment that addresses these gaps through one connected workflow.

The main objectives are:

- provide a unified environment for industrial and educational robot simulation;
- integrate FK, IK, Jacobian, trajectory, and workspace analysis with the simulator;
- connect mathematical robot calculations directly to interactive 3-D visualization; and
- support structured practical challenges that allow users to apply robotics concepts through simulation.

III. RELATED WORK

The three studies listed in the project presentation were selected because they directly relate to web-based robotics simulation and learning. Table I summarizes the role of each study in motivating the project.

The present project follows the same broad educational direction but places more emphasis on integrating a larger serial-robot catalog, multiple analysis modules, real-time 3-D interaction, and practical challenge workflows in a single environment.

IV. SYSTEM ARCHITECTURE AND METHODOLOGY

A. Overall Architecture

The application is organized around a common robot state. A selected robot definition provides its joint types, DH parameters, joint limits, home configuration, and visual model information. The active joint configuration is then processed by the kinematics layer to produce transformations, end-effector pose, Jacobian information, trajectory state, and other analysis

TABLE I
THREE REFERENCE WORKS USED IN THE PROJECT

Reference	Main focus	Relevance to KyntriX
Dey and Cheruvu [1]	WebGL modelling and kinematics	Demonstrates browser-based 3-D robot modelling, FK/IK study, and CAD/STL integration.
Peidr� et al. [2]	Mobile virtual laboratory	Shows the educational value of visual, browser-accessible kinematic experimentation.
Nandi and Pattanaik [3]	Web-based robotics simulator	Demonstrates browser-based modelling, kinematics, and motion-oriented learning.

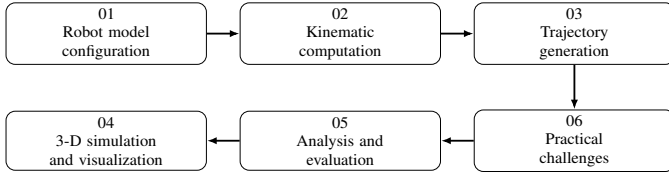


Fig. 1. Six-stage KyntriX workflow used to connect modelling, computation, simulation, analysis, and practice.

quantities. These results are reused by the simulator, analysis views, trajectory tools, and learning components.

The central idea is that the rendered model should represent the same computed state that appears in the mathematical views. A change in joint value therefore propagates through the kinematic computation and is reflected in the 3-D scene, end-effector pose, trace, and related analysis.

B. Robot Model Configuration

The catalog contains 42 supported robot configurations, divided into 22 industrial models and 20 educational skeletons. The industrial side includes representative ABB, FANUC, KUKA, PUMA, Epson, Stanford, and tabletop models, while the skeleton set provides simpler serial chains for academic analysis. A custom robot creator is also available for user-defined serial manipulators.

Each robot definition stores structured joint information, including joint type, DH parameters, limits, home position, and visual model information. This allows the simulator and analysis modules to operate on the same definition. The custom creator supports one to eight degrees of freedom, revolute or prismatic joints, Standard or Modified DH selection, joint limits, and JSON import/export.

V. CORE USER-FACING MODULES

The project presentation emphasizes a workflow that is visible to the user rather than only an internal numerical engine. The main modules are summarized in Table II.

TABLE II
CORE USER-FACING KYNTRIX MODULES

Module	User-facing capability
Robot Model Configuration	Select industrial or educational robots, inspect DH data, joint limits, link geometry, and create custom serial manipulators.
DH and Transformation Tools	Edit or inspect DH parameters, view individual and cumulative homogeneous transformations, and observe coordinate-frame changes in 3-D.
Forward Kinematics	Move joints and obtain the resulting end-effector position, orientation, and transformation chain.
Inverse Kinematics	Set a target pose and solve for a valid joint configuration using suitable analytical or numerical methods.
Jacobian and Dexterity Analysis	Inspect Jacobian-based motion relationships, singularity-related information, manipulability, and condition measures.
Trajectory and Motion Analysis	Generate joint-space or Cartesian motion and inspect position, velocity, acceleration, end-effector motion, and trajectory behaviour.
3-D Simulation and Practical Learning	Interact with the robot in real time, trace the end-effector, explore workspace behaviour, and use the same simulator in Problem Arena and pose challenges.

A. DH and Transformation Tools

DH modelling provides the geometric basis for the serial manipulator. For Standard DH, the link transform is

$$A_i = R_z(\theta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i), \quad (1)$$

where θ_i is the joint angle, d_i is the link offset, a_i is the link length, and α_i is the link twist. The cumulative transformation is

$$T_{0i} = A_1 A_2 \cdots A_i. \quad (2)$$

The resulting matrix contains the end-effector position and orientation. Modified DH is also supported, allowing the user to select the convention appropriate to the robot or exercise being studied.

The transformation view makes the mathematics visible without requiring the learner to leave the simulator. Coordinate frames can be inspected alongside the robot, so a parameter change is related directly to the corresponding geometric change.

B. Forward and Inverse Kinematics

Forward kinematics maps a joint configuration to the end-effector pose. In KyntriX, the FK result is shared by the simulation and analysis layers, so the displayed robot, matrix values, and end-effector readout remain connected.

Inverse kinematics solves the reverse problem: finding joint values that achieve a target pose. The project contains analytical solvers for suitable structures and a numerical damped least-squares pathway for cases where a closed-form solution is not directly applicable. Returned solutions are evaluated against the forward-kinematics model and joint limits before being treated as valid results. This keeps the practical workflow tied to the same mathematical model used for visualization.

C. Jacobian, Singularity and Dexterity Analysis

The Jacobian relates changes in joint variables to end-effector motion. KyntriX exposes this information through analysis

views rather than treating it as an isolated numerical calculation. The interface can show Jacobian-related rank and conditioning measures together with manipulability information, allowing the user to study configurations in which robot motion becomes less favourable or loses degrees of freedom.

The purpose is primarily educational and analytical: the user can compare a normal configuration with a singularity-prone configuration and see how the corresponding robot state changes. The project does not claim that numerical indicators alone provide a complete physical safety analysis.

D. Trajectory Generation and Workspace Exploration

The trajectory module supports joint-space motion and Cartesian motion. Joint trajectories are generated from start and goal configurations, while Cartesian trajectories interpolate end-effector position and orientation and track the resulting path through inverse kinematics. The implementation also evaluates the kinematic state at the sampled trajectory points so that motion and analysis remain synchronized.

Workspace exploration provides sampled information about reachable and dexterous regions. The project presentation explicitly identifies workspace analysis as part of the integrated robotics computation workflow. The current implementation treats these boundaries as sampled computational results rather than complete representations of an industrial workcell.

E. 3-D Simulation and Visualization

The simulation stage presents the active robot model as an interactive three-dimensional scene. Users can select robot models, modify joints, inspect coordinate frames, observe the end-effector path, control the camera, and interact with target poses. The robot geometry is linked to the active kinematic state rather than being animated independently of the mathematics.

This relationship is important for learning. A student can change a joint in the control interface, inspect the corresponding transformation matrix, watch the link move in 3-D, and then study the effect on the end-effector or trajectory. The same state can subsequently be used by the learning and challenge modules.

VI. PRACTICAL LEARNING: PROBLEM ARENA AND CHALLENGES

The learning component extends the simulator beyond passive visualization. The Problem Arena covers core robotics concepts such as forward kinematics, inverse kinematics, DH parameters, and singularity or dexterity reasoning. Problems can use answers such as scalars, angles, vectors, matrices, DH tables, and joint configurations. The interface also provides progressive hints and connects selected problems to the simulator.

Simulation challenges provide a second practical layer. A challenge can specify a robot and a target pose, after which the learner can load the target into the 3-D scene, modify the joints or use IK, and submit the resulting configuration for evaluation. The scoring process checks pose error against the challenge

TABLE III
FUNCTIONAL IMPLEMENTATION STRUCTURE

Layer	Main responsibility
Robot data	Stores robot definitions, joint types, DH parameters, limits, home configurations, and references to 3-D model assets.
Robotics computation	Provides DH transformations, forward kinematics, inverse kinematics, Jacobian and dexterity calculations, and trajectory evaluation.
Simulation	Builds the interactive 3-D scene, applies the computed joint configuration to robot links, displays frames, traces motion, and provides camera interaction.
Application state	Maintains the active robot, joint values, selected analysis mode, targets, and simulation state so that different views operate on the same configuration.
Learning and evaluation	Provides Problem Arena activities, simulator-linked exercises, pose challenges, attempts, and deterministic evaluation of submitted configurations.
Validation and testing	Checks mathematical functions and integrated behaviour, including catalog coverage, FK/IK behaviour, trajectories, simulation interactions, and learning workflows.

tolerances, so the exercise is based on the robot's actual kinematic state rather than on a manually judged screenshot.

The practical value of this approach is the transition from theory to spatial reasoning. A learner may first calculate a kinematic quantity, inspect the corresponding robot state in 3-D, and then solve a target-pose exercise using the same model. This directly supports the project objective of learning through practice while keeping the interaction within one environment.

VII. IMPLEMENTATION STRUCTURE AND INTERACTION FLOW

The user-facing behaviour is supported by a modular implementation that separates robot data, mathematical computation, simulation, application state, and learning features. This separation is useful for the project because a change in one part of the application should not require a second, unrelated implementation of the same robotics quantity. The report deliberately describes this structure at a functional level so that the important design decisions remain clear without exposing unnecessary implementation detail.

A. Robot Data to 3-D Scene

The workflow begins with a structured robot definition rather than a manually assembled visual scene. Selecting a robot therefore establishes both the mathematical description and the corresponding geometry. The definition contains the information needed to identify the number and type of joints, the DH parameters used to compute their transforms, their allowable ranges, and the model assets required for visualization.

Once a joint configuration is active, the forward-kinematics computation produces a sequence of link transforms and the final end-effector pose. The simulation stage consumes these transforms to place the robot links and coordinate frames. This arrangement makes the 3-D model a visual representation of the calculated state. The user can consequently inspect a matrix, look at the associated coordinate frame, and then observe the same frame relationship in the rendered robot.

B. Interactive Analysis Flow

A typical analysis session follows a short sequence. First, the user selects an industrial or educational robot, or creates a custom serial chain. Second, the DH table and joint limits are inspected or edited where the selected module permits editing. Third, the current joint configuration is evaluated to obtain the transformation chain and end-effector pose. Fourth, the user may request an inverse-kinematics solution for a target pose, generate a trajectory, or examine Jacobian and workspace information. Finally, the resulting state can be explored directly in the 3-D scene or transferred into a practical challenge.

This sequence is useful because each step reuses the previous one. A target pose does not create a separate visual robot state; it is solved using the robot already selected in the simulator. Likewise, a trajectory is not merely a plotted curve. Its sampled joint configurations are evaluated through the same kinematic model and then displayed in the scene. The result is a consistent path from mathematical input to visible motion.

C. Role of the Shared State

The shared-state principle is especially important when several modules are open. A joint value changed through the control interface should agree with the value shown in the analysis view, the value used by FK, and the configuration displayed by the simulator. Similarly, when a challenge is loaded into the simulator, the target pose and robot identity should remain tied to that challenge throughout the interaction.

This design also supports progressive learning. Beginners can work with simple skeletons and inspect one transformation at a time. More advanced users can move to six-degree-of-freedom industrial models, compare multiple IK branches, examine Jacobian behaviour, or follow a trajectory through the workspace. The custom robot creator extends the same state model to user-defined examples, allowing experimentation without changing the basic workflow.

D. Scope of the Numerical Pipeline

The numerical pipeline is intentionally focused on the quantities required by the project. It covers rigid serial-chain kinematics, pose computation, inverse-kinematics solving, Jacobian-based analysis, trajectory sampling, and workspace exploration. The implementation does not attempt to model complete rigid-body dynamics, certified actuator behaviour, physical contact, or an industrial robot cell. Keeping the scope explicit is important because the application is intended for simulation, analysis, and education rather than for direct control of physical machinery.

VIII. VALIDATION AND RESULTS

The repository contains validation at both the component level and the catalog level. Tests cover DH transformations, Standard and Modified DH behaviour, forward kinematics, Jacobian calculations, analytical and numerical inverse kinematics, trajectories, simulation behaviour, and learning workflows. The catalog-level validation checks 42 distinct models, their joint definitions and limits, referenced industrial mesh assets,

TABLE IV
DOCUMENTED SKELETON 6R AUDIT RESULTS

Check	Cases	Passed	Failed	Failure Rate
FK matrix comparison	15	15	0	0.00%
IK pose validation	15	15	0	0.00%
Joint-limit validity	15	15	0	0.00%

FK against an independent Standard-DH calculation, Jacobian metrics, and IK round-trip behaviour.

A separate documented audit provides a more focused verification of the Skeleton 6R model. Fifteen fixed joint configurations were checked in two directions. First, application FK was compared element-by-element with an independent Standard-DH implementation. Second, target poses were generated independently and passed through the application IK pipeline, after which the returned solutions were checked again through independent FK.

The same audit reports a maximum IK position residual of 1.759×10^{-6} m and a maximum orientation residual of 2.107×10^{-8} rad for the tested cases. These values are evidence for the selected model and test set; they are not treated as a universal proof for every robot state, singular configuration, or unreachable target.

The project presentation identifies three practical outcomes: unified robot simulation, integrated robotics computation, and a working web-based laboratory for simulation, analysis, and practical robotics challenges. In the implemented environment, these outcomes are visible as one user workflow rather than separate demonstrations. A robot is selected or created, its kinematic model is inspected, its pose is computed and visualized, motion can be generated and analysed, and the same state can be used in a structured challenge.

IX. REPRESENTATIVE USER SCENARIOS AND OUTCOMES

The integrated design supports several common robotics-learning scenarios without changing the underlying workflow. A beginner can start with an educational skeleton, inspect a small DH table, move one joint, and observe the corresponding frame and end-effector change. A student working with an industrial model can then repeat the same process with a six-degree-of-freedom manipulator and study the additional orientation and inverse-kinematics behaviour. In both cases, the interface remains based on the same robot state.

An important outcome of these scenarios is that the learner is not forced to switch between unrelated representations. The DH table, matrices, joint controls, trajectory states, and 3-D model refer to the same robot configuration. This makes the platform suitable for demonstrations in which an instructor changes a parameter and immediately asks students to predict the resulting motion, as well as for individual practice in which the learner explores a robot independently.

The same workflow also supports comparison. For example, a learner can compare an educational 3R or SCARA structure with a six-degree-of-freedom industrial arm, observe how their available motion differs, and connect those differences to their

TABLE V
REPRESENTATIVE USE OF THE PLATFORM

Scenario	Typical interaction and outcome
DH learning	Inspect or edit parameters, view frames, and observe the geometric effect of changing a link value or joint variable.
FK study	Enter or jog a joint configuration and inspect the resulting transformation chain and end-effector pose.
IK study	Provide a target pose, solve for a reachable joint configuration, and verify the resulting pose in 3-D.
Motion study	Select start and goal states, generate a trajectory, and inspect position, velocity, acceleration, and end-effector motion.
Robot design	Build a custom 1–8 DOF serial manipulator and explore its kinematics using the same simulation workflow.
Practical challenge	Load a target, use the simulator to reach it, and submit the resulting configuration for tolerance-based evaluation.

kinematic models. Similarly, a normal configuration can be compared with a singularity-prone configuration to study the meaning of Jacobian rank and conditioning. These interactions strengthen the connection between equations, geometry, and observed robot behaviour.

From an evaluation perspective, the platform also gives a clearer boundary between simulation output and practical assessment. A challenge can be defined using a robot, target pose, and error tolerances, while the learner’s answer is evaluated from the same kinematic model that drives the scene. This is more consistent than judging a screen image alone because the submitted joint configuration can be checked numerically.

The overall outcome is therefore a single laboratory-style workflow: configure, calculate, visualize, analyse, and practice. That sequence captures the central idea of KyntriX—simulation is not presented as an isolated animation, but as a way to connect formal robotics concepts with interactive experimentation.

X. LIMITATIONS AND FUTURE SCOPE

The current system has explicit boundaries. It does not provide physical robot integration, collision detection is not yet implemented, workspace boundaries are sampled, and browser-based computation has practical limits for larger workloads. These limitations are consistent with the project’s scope as a simulation and learning laboratory rather than a certified industrial controller.

The identified future scope includes real-robot hardware integration, robot and environment collision detection, more advanced workspace computation, and improvements in computation speed and scalability. These extensions would increase the realism and usefulness of the environment while keeping the existing kinematic foundation intact.

XI. DISCUSSION AND EDUCATIONAL SIGNIFICANCE

The main contribution of KyntriX is the continuity of its workflow. A DH table is connected to coordinate frames, the coordinate frames are connected to the transformation chain, the transformation chain is connected to the end-effector pose, and the pose is connected to the 3-D robot. The same robot state can then drive trajectory generation and practical exercises.

This reduces the conceptual gap between a formula written on paper and the movement observed in the simulator.

The breadth of the catalog also supports different levels of learning. Educational skeletons can be used for basic joint and DH exercises, while industrial models such as PUMA, ABB, KUKA, FANUC, and SCARA examples can be used for more realistic manipulation scenarios. The custom creator extends the same workflow to user-defined serial chains, allowing students to experiment with their own kinematic structures rather than only predefined examples.

The platform is therefore best understood as a robotics laboratory rather than only a visual simulator. Its emphasis is on learning through interaction: configure a model, compute its state, observe the state in 3-D, analyse the motion, and apply the result in a practical task. That integrated experience is the central purpose of the project.

XII. CONCLUSION

KyntriX brings robot simulation and kinematic analysis together in a single browser-based environment. The project combines 42 supported robot configurations with DH modelling, FK and IK, Jacobian and singularity analysis, trajectory generation, workspace exploration, interactive 3-D visualization, custom robot creation, and simulator-linked practical challenges. Its shared-state approach keeps mathematical results and rendered motion connected throughout the workflow. The documented validation process provides supporting evidence for the implemented kinematic pipeline while clearly identifying the limits of the current test coverage. With future additions such as collision detection and physical robot integration, the same architecture can be extended while preserving its core purpose as a practical environment for learning, testing, and experimenting with robotics.

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